Name ____ Date ___________ Period _____

Domain and Range from Problem Situations

TEACHER KEY | Algebra II | Unit 1 | Day 2 | TEKS A2.7.I, A2.2.A

Objective: I can find the domain and range of a real-world situation, explain what they mean in context, and tell the difference between the domain of a function and the domain that the situation allows.

Vocabulary

Independent variable — the **input**, the one you choose. Goes on the **x**-axis.

Dependent variable — the **output**, it depends on the input. Goes on the **y**-axis.

Function domain — every input the **equation** will accept.

Contextual domain — every input the **situation** actually allows. Usually **smaller**.

Part 1 — Name the Variables

Which quantity is the input and which is the output?

1. Hours you work and the money you earn
   independent: **hours worked** dependent: **money earned**

2. Number of tickets bought and the total cost
   independent: **tickets bought** dependent: **total cost**

3. Seconds since a ball was thrown and its height
   independent: **seconds** dependent: **height**

Part 2 — Function Domain vs. Contextual Domain

A cube-shaped box has side length x inches. Its volume is $\mathbf{V(x) = x^3}$.

What domain does the **equation** x^3 accept? **all real numbers**

What domain does the **box** allow? **$x > 0$, or $(0, \infty)$**

Why? **a side length cannot be zero or negative**

Big idea: the equation may accept numbers the **situation** does not. Always ask what makes sense.

Part 3 — Domain and Range in Context

Give the contextual domain and range. Use an inequality and interval notation.

Example 1. Concert tickets cost $12 each. You may buy at most 8. Total cost $C(t) = 12t$.
Domain: **$t = 0, 1, 2, \ldots 8$ (whole tickets only)** Discrete or continuous? **discrete**

Range: **$\{ 0, 12, 24, 36, 48, 60, 72, 84, 96 \}$**

You Try 1. A ball is thrown. Its height is $h(t) = -16t^2 + 64t$ feet, and it is in the air from $t = 0$ to $t = 4$ seconds. Its greatest height is 64 feet.
Domain: **$0 \le t \le 4$** interval **$[0, 4]$** Discrete or continuous? **continuous**

Range: **$0 \le h \le 64$** interval **$[0, 64]$**

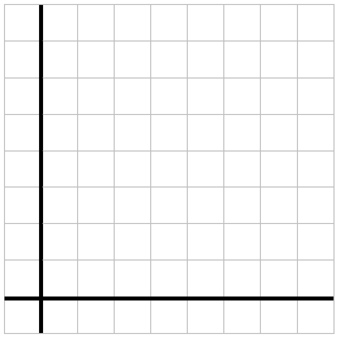
You Try 2. A parking garage charges $3 per hour and is open 6 a.m. to 10 p.m. (16 hours).
Domain: **$0 \le h \le 16$, $[0, 16]$** Range: **$0 \le c \le 48$, $[0, 48]$**

Part 4 — Reading a Situation from a Graph

Each square is 1 unit. Plot the points, then answer in context.

A water tank is drained. Gallons g after m minutes:

m	0	1	2	3	4
g	8	6	4	2	0



Answer in context.

Domain: **$0 \le m \le 4$, [0, 4]**

Range: **$0 \le g \le 8$, [0, 8]**

What does the domain mean here?

the tank drains over 4 minutes

What does a range of 0 mean?

the tank is empty

Part 5 — Does the Situation Restrict the Domain?

Write yes or no, and if yes, give the restriction.

1. The side length of a square garden **yes, $x > 0$**
2. The temperature in a freezer over one day **no, it may be negative**
3. The number of buses needed for a field trip **yes, whole numbers only**

Practice

Give the contextual domain and range for each. Show your thinking on the lines.

1. A phone plan costs $30 per month plus $5 per gigabyte, up to 20 GB.

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Domain **$0 \le g \le 20$, [0, 20]** Range **$30 \le c \le 130$, [30, 130]**

2. A theater has 250 seats. Tickets are $9 each, and t tickets are sold.

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Domain **t = 0, 1, 2, ... 250** Range **{ 0, 9, 18, ... 2250 }**

3. A diver descends from the surface to 40 meters below sea level.

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Range of depth **$-40 \le d \le 0$, [-40, 0]** Why negative? **below sea level**

Exit Ticket

A taxi charges a $4 flat fee plus $2 per mile, for trips up to 10 miles. Cost C(m) = 4 + 2m.

Domain **$0 \le m \le 10$, [0, 10]** Range **$4 \le C \le 24$, [4, 24]**

Why does the range start at 4, not 0? **the $4 flat fee is charged even at 0 miles**